

Article

A Combined System Metrics Approach to Cloud Service Reliability Using Artificial Intelligence

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Abstract: Identifying and anticipating potential failures in the cloud is an effective method for increasing cloud reliability and proactive failure management. Many studies have been conducted to predict potential failure, but none have combined SMART (self-monitoring, analysis, and reporting technology) hard drive metrics with other system metrics, such as central processing unit (CPU) utilisation. Therefore, we propose a combined system metrics approach for failure prediction based on artificial intelligence to improve reliability. We tested over 100 cloud servers' data and four artificial intelligence algorithms: random forest, gradient boosting, long short-term memory, and gated recurrent unit, and also performed correlation analysis. Our correlation analysis sheds light on the relationships that exist between system metrics and failure, and the experimental results demonstrate the advantages of combining system metrics, outperforming the state-of-the-art.

Keywords: failure prediction; fault tolerance; cloud computing; artificial intelligence; reliability



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1. Introduction

Cloud computing has emerged as the fifth utility over the last decade, and is a backbone to the modern economy [1]. It is a model of computing that allows flexible use of virtual servers, massive scalability, and management services for the delivery of information services. With the low-cost pay-per-use model of on-demand computing [2], the cloud has grown massively over the years, both in terms of size and complexity.

Today, almost everyone is connected to the cloud in one way or another. This is because of cost effectiveness with a pay-as-you-go or subscription-based service model for on-demand access to IT resources [1,2]. Industries rely on the cloud for their operations, academicians to accelerate and conduct scientific experiments, and ordinary end-users by using cloud-based services knowingly or unknowingly, such as Google Drive, Gmail, Outlook, and so on. Furthermore, the cloud today is more important than yesterday, as it supports smart city construction [3], enterprise business [4], scalable data analysis [5,6], healthcare [7,8] and also new evolving computing paradigms, such as fog and edge computing [9].

To date, despite the significant improvement in the performance of the hardware elements of the cloud infrastructure, the failure rate remains substantial. Moreover, the cloud is not as reliable as the cloud service providers, such as Amazon AWS and Ali Cloud, claimed, which is more than 99.9% [10]. For example, multiple instances of failure have

precision and F1 score. Further, our combined system approach demonstrates competitive results by nearly 5% in terms of F1 score against state-of-the-art studies and precision by up to 10%. The studies FACS and DESH used LSTM as in our study. Based on the findings of these two studies, our proposed combined system approach continues to outperform the competition. However, when we compare our LSTM and GRU (i.e., a similar technique) results to the maximum DESH result, we see that DESH outperforms our proposed combined system metrics in terms of precision by up to 3%.

Finally, the results from our experiment and comparison to state-of-the-art studies substantiate our claim that using SMART metrics in conjunction with other system metrics, such as CPU and memory utilisation, can improve failure prediction. Furthermore, with improved failure prediction, we can detect potential failures accurately before they occur, allowing us to take proactive action and, as a result, improve reliability.

5. Conclusions

We hypothesised that combining multiple system metrics would improve failure prediction, and in this paper, we present our research on cloud server failure prediction. Additionally, we substantiated our hypothesis and demonstrated the competitive advantage of combined metrics through data analysis, a correlation study, and experimental results based on our acquired data from 100 different cloud servers. As a result of the data analysis and correlation study, we observed the following: (i) SMART system metrics have a very low, if any, correlation with other system metrics, such as CPU utilisation; (ii) in the event of a failure, high resource utilisation can be observed; and (iii) the benefits of combining system metrics. For instance, SMART system metrics do not include failure information, whereas other system metrics do (see Section 4.4.1). Subsequently, in a comparison of our automated failure prediction models, we discovered that all four AI algorithms, RF, GB, LSTM, and GRU, that utilise combined system metrics (or multi-modal inputs) outperformed the state-of-the-art studies. The highest accuracy, however, was obtained by RF methods, compared to other alternatives. Moreover, for competitive reasons, cloud providers and high-performance computing centres do not want their failures to be made public. This is also why such failure datasets are scarce, and those that exist are extremely old. Utilising such data (i.e., very old data) will not provide the necessary insights, for example, as a result of hardware (or technological) advancements. As a result, we were required to generate the target label, a procedure similar to that used in other studies (see Section 3.3). This, however, may not accurately reflect the actual failure pattern, as failures can occasionally occur abruptly, which we regard as a limitation of this study. The recent work by Chhetri et al. [100] on failure prediction, which focuses on hard drive failure prediction using knowledge graphs [101], has demonstrated even greater effectiveness. Future work will involve expanding on this approach of combining system metrics based on knowledge graphs, as described in [100], in order to achieve even greater effectiveness, as well as implementing the production method.

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